- Reference ??

Integer Modulo

Modulo **modulo** **mod** a **mod** q **(i.e.,** $a modulo q$**)**

a q $\{0, 1, 2, 3, \cdots, q-1\}$ $7 \bmod 5 = 2$

Modulus: a q q *modulo*

Modulo Congruence ($\equiv$): a b q $a \equiv b$ **mod** q q $5 \equiv 12 \bmod 7$ $5 \bmod 7 = 5$ $12 \bmod 7 = 5$ $a \equiv b \bmod q$ $a = b \pmod q$ a q b q $a = b \bmod q$ a b q

Congruence *vs.* Equality: $a \equiv b \bmod q \Longleftrightarrow a = b + k \cdot q$ k a b q a b q $5 \equiv 12 \bmod 7 \Longleftrightarrow 5 = 12 + (-1) \cdot 7$

??.1 Properties of Modulo Operations

x

Addition: $a \equiv b \bmod q \Longleftrightarrow a + x \equiv b + x \bmod q$

Subtraction: $a \equiv b \bmod q \Longleftrightarrow$